

This session, as the first one, aims at taking in hand some mathematical tools available to describe random elements. A set of various exercises are proposed. An numerical study will be carried out using Matlab to realize the information contained in the different mathematical entities, and their respective limits.

1 Cumulative distribution function and probability density function

- (1). Create a function in **Matlab** that reads a data set and plots the cumulative distribution function (CDF) from an experimental set of discrete data. Compare this function with the Matlab functions **ecdf**.
- (2). Estimate the probability density function (PDF) from an experimental set of data.
- (3). Estimate the cumulative distribution function from the probability density function.

2 Covariance and correlation

The covariance of two random variables is calculated by:

$$\text{Cov}(X, Y) = \mathbb{E}\{(X - \mathbb{E}\{X\})(Y - \mathbb{E}\{Y\})\}. \quad (1)$$

- (1). Implement a function which computes the covariance of two variables. Compare with the function **cov**.
- (2). Implement a function which computes the correlation between two variables. Compare with **corrcoef**.

Consider the random variables:

- r : the discrete random variable describing the number of cases,
 - u : the continuous random variable describing the mean mass of the cases with a uniform distribution,
 - n : the continuous random variable describing the mean mass of the cases with a normal distribution,
- (3). Using the function previously implemented or the Matlab function **cov**, investigate the covariance:
 - between the different random variables as previously defined,
 - between one of this random variable and itself,

Consider a random variable x uniformly distributed on $[-10; 10]$ and $y = x^2$, where y is completely determined by x , therefore x and y are not independent, but their correlation - as their covariance - is 0.

- (4). Which properties exhibits this matrix?
- (5). Using the function previously implemented or the Matlab function **corrcoef**, investigate the correlations between the different random variables (r, u, n) .

3 Random vectors

- (1). Simulate a random vector:
 - Choose a sample space and a probability distribution of the random experiment.
 - Simulate $n_1 = 10$, and $n_2 = 100$ numerical experiments of the vector.
 - Estimate the expected value and the covariance matrix of the random vector. How is this covariance matrix computed? (Hint: Use the **reshape** function to reorder the matrix containing the random variable in vectors). Compare results with the Matlab built-in function **cov(n, 2*n, 1)**.

4 Random fields

- (1). Plot one realization of a random field, e.g. along the length of a beam $x_0 = [0, 5]$ m, and the random variable relative to one position x_0 for:
 - a one-variate one-dimensional random field,
 - a two-variate one-dimensional random field.
 - a one-variate two-dimensional random field.
- (2). Choose some values of the random field, and estimate the values of the autocorrelation function of this random field for the relative points.
- (3). Investigate the autocorrelation function $C_{HH}(x_1, x_2) = \exp -\frac{|x_1 - x_2|}{b}$. How does the function evolve with respect to the parameter b ? What is the name of this parameter?

5 Two random variables

Consider the bivariate Gaussian probability distribution for Young's modulus E and Poisson's ratio ν . They have respectively the mean values μ_E and μ_ν , and the standard deviations σ_E and σ_ν .

The bivariate Gaussian probability density function $f_{E\nu}$ reads:

$$f_{E\nu}(E, \nu) = \frac{1}{2\pi\sigma_E\sigma_\nu\sqrt{1-\rho_{E\nu}^2}} \exp -\frac{1}{2(1-\rho_{E\nu}^2)} \left[\frac{(E-\mu_E)^2}{\sigma_E^2} + \frac{(\nu-\mu_\nu)^2}{\sigma_\nu^2} - \frac{2\rho_{E\nu}(E-\mu_E)(\nu-\mu_\nu)}{\sigma_E\sigma_\nu} \right] \quad (2)$$

where $\rho_{E\nu}$ is the correlation between the two random variables E and ν .

We consider the numerical application: $\mu_E = 21\,000$ MPa, $\sigma_E = 1050$ MPa, $\mu_\nu = 0.3$, $\sigma_\nu = 0.03$ and $\rho_{E\nu} = 0.20$.

- (1). Represent the joint probability density function $f_{E\nu}$. Does this function satisfy the condition:

$$\int_{\omega_X \omega_Y} f_{XY} \, dx \, dy = 1?$$

- (2). Represent the marginal probability density function of ν . Estimate the probability that $\nu = 0.25$.
- (3). Represent the marginal probability density function of E . Estimate the probability that $E = 21000$ MPa.
- (4). Estimate the probability that $E = 21000$ MPa while $\nu = 0.25$.
- (5). Estimate the probability that $E \leq 21000$ MPa while $\nu \leq 0.25$.
- (6). Are the two random variables E and ν independent?
- (7). Plot the conditional probability of E , given $\nu = \bar{\nu}$. Compare the probability density function of E to the probability density function of E given $\nu = 0.3$, for $\rho_{E\nu} = 0.20$, $\rho_{E\nu} = 0.40$ and $\rho_{E\nu} = 0.00$.